ROYAL GLOBAL UNIVERSITY
GUWAHATI

(ROYAL SCHOOL OF MEDICAL & ALLIED SCIENCES)

(RSMAS)

DEPARTMENT OF DIALYSIS THERAPY TECHNOLOGY

Course Sstructure & Syllabus

In Line with National Commission for Allied and Healthcare Professions (NCAHP)

For

MASTER OF DIALYSIS THERAPY

W.E.F 2026 – 27

Curriculum

MASTER OF DIALYSIS THERAPY

(2 years program)

Curriculum Schemes

Semester I														
Course Code	Course	Result in Group (RG) #	Credit/ Week				Hours/ Semester			Attendance %	Continuous Internal	End Semester		Grand Total
			Lecture	Tutorial/ Clinical	Practical/	Credits	Lecture	Practical/	Total Hours			Theory (b)	Practical/ Viva	Theory: a+(b÷2)
MDT244C101	Anatomy (Basic anatomy/ Nephroanatomy / Embryology/ Histology)	A	4			4	60		60	80	50	100		100
MDT244C102	Physiology (General physiology/ Nephrophysiology/ Endocrinology)	B	4			4	60		60	80	50	100		100
MDT244C103	Biochemistry	C	4			4	60		60	80	50	100		100
MDT244C104	Microbiology	D	4			4	60		60	80	50	100		100
MDT244C105	Renal Genetics & Molecular Biology		4			4	60		60	80	50	100		100
MDT244C111	Anatomy (Practical)	A			2	2		60	60	80	50		50	100
MDT244C112	Physiology (Practical)	B			2	2		60	60	80	50		50	100
MDT244C113	Biochemistry (Practical)	C			2	2		60	60	80	50		50	100
MDT244C114	Microbiology (Practical)	D			2	2		60	60	80	50		50	100

MDT244C115	Residency - 1			2		2		90	90	80	50		50	100
Total			20	2	8	30	300	330	630		500	500	250	1000

Semester II														
Course Code	Course	Result in Group (RG) #	Credit/ Week				Hours/ Semester			Attendance %	Continuous Internal	End Semester		Grand Total
			Lecture	Tutorial/ Clinical Training	Practical/ Research Project	Credits	Lecture	Practical/ Tutorial	Total Hours			Theory (b)	Practical/ Viva (c)	Theory: a+(b÷2)
MDT244C201	Renal Pathology	E	4	-	-	4	60	-	60	80	50	100	-	100
MDT244C202	Pharmacology	F	4	-	-	4	60	-	60	80	50	100	-	100
MDT244C203	Nephro-Immunology	G	3	-	-	3	45	-	45	80	50	100	-	100
MDT244C204	Research Methodology and Biostatistics		4	-	-	4	60	-	60	80	50	100	-	100
MDT244C241	Biophysics & Biomedical Instrumentation		2	-	1	3	30	30	60	80	50	100	-	100
MDT244C205	Elective Course (To be chosen by		3	-	-	3	45	-	45	80	50	100	-	100

	the student)													
MDT244C211	Renal Pathology (Practical)	E	-	-	1	1	-	30	30	80	50	-	50	100
MDT244C212	Pharmacology (Practical)	F	-	-	1	1	-	30	30	80	50	-	50	100
MDT244C213	Nephro-Immunology (Practical)	G	-	-	1	1	-	30	30	80	50	-	50	100
MDT244C214	Residency - 2		-	3	-	3	-	135	135	80	100	-	50	100
Total			20	3	4	27	300	255	555		500	600	200	1000

Semester III														
Course Code	Course	Result in Group (RG) #	Credit/ Week				Hours/ Semester			Attendance %	Continuous Internal	End Semester		Grand Total
			Lecture	Tutorial/ Clinical Training	Practical/ Research Project	Credits	Lecture	Practical/ Tutorial	Total Hours			Theory (b)	Practical/ Viva (c)	Theory: a+(b÷2)
MDT244C301	Renal Disease	H	4	-	-	4	60	-	60	80	50	100	-	100
MDT244C302	Emergency Medicine and Renal Nutrition& Dietetics	I	3	-	-	3	45	-	45	80	50	100	-	100
MDT244C303	Nephro-Radiology and Imaging sciences	J	4	-	-	4	60	-	60	80	50	100	-	100
MDT244C311	Renal Disease (Practical)	H	-	-	1	1	-	30	30	80	50	-	50	100
MDT244C312	Emergency Medicine and Renal Nutrition & Dietetics (Practical)	I	-	-	1	1	-	30	30	80	50	-	50	100
MDT244C313	Nephro-Radiology and Imaging sciences (Practical)	J	-	-	1	1	-	30	30	80	50	-	50	100
MDT244C315	Residency - 3		-	5	-	5	-	225	225	80	100	-	-	100

MDT244C331	Seminar on Dissertation		-	-	3	3	-	90	90	80	100	-	-	100
Total			11	5	6	22	165	405	570		500	300	150	800

Semester IV														
Course Code	Course	Result in Group (RG) #	Credit/ Week				Hours/ Semester			Attendance %	Continuous Internal	End Semester		Grand Total
			Lecture	Tutorial/ Clinical	Practical/	Credits	Lecture	Practical/	Total Hours			Theory (b)	Practical/ Viva	Theory: a+(b÷2)
MDT244C401	Advanced Renal Replacement Therapy	K	4	-	-	4	60	-	60	80	50	100	-	100
MDT244C402	Renal transplantation and coordination/Psychology	L	4	-	-	4	60	-	60	80	50	100	-	100
MDT244C403	Quality assurance and management of Dialysis unit	M	4	-	-	4	60	-	60	80	50	100	-	100
MDT244C411	Advanced Renal Replacement Therapy (Practical)	K	-	-	1	1	-	30	30	80	50	-	50	100
MDT244C412	Renal transplantation and	L	-	-	1	1	-	30	30	80	50	-	50	100

	coordination/Psychology (Practicals)													
MDT244C413	Quality assurance and management of Dialysis unit (Practical)	M	-	-	1	1	-	30	30	80	50	-	50	100
MDT244C414	Residency - 4		-	4	-	4	-	180	180	80	100	-	-	100
MDT244C421	Dissertation and viva voce		-	-	6	6	-	180	180	80	50	-	50	100
Total			12	4	9	25	180	450	630		450	300	200	800
Grand Total			63	14	27	104	945	1440	2385		1950	1700	800	3600

Results in Group (RG) - Programs wherein Theory and Practical's (Lab Course) are assessed jointly (RG), the passing minimum for the theory exams and Practical exams have to be obtained separately, in order to be declared passed in the individual courses and thereby earning the credits for both Theory and Practical courses. Reappearance in any one of the components(Theory or Practical) is treated as reappear in both these components (Theory and Practical's).

First Semester

Anatomy (60 Hours)

1. Anatomical Terminology: Descriptive anatomical terms, terms related to movement and position.

2. Skeletal system-Name and parts of the Bone and joints

a. Skeleton, skull, vertebrae, pelvis, ribs, principal joints-Hip joint, Shoulder joint, Sacroiliac joint.

b. Lumbar vertebra and pelvis in detail

3. Muscular system – Classification, Types, Attachment, Action and nerve supply of the Muscles of Abdomen and Pelvis

4. Cardiovascular system – Heart and Major Blood Vessels- arteries of upper limb, lower limb, -Arch of Aorta and its Branches Abdominal Aorta and its branches, Inferior Vena cava and its Tributaries, Pericardium& Pericardial sinuses.

5. Heart: -External and Internal features

a. Structure of the heart

b. Conducting system of the heart,

c. Blood supply of the Heart

d. Nervous supply of the heart

e. Valves of the heart

6. Respiratory system – Pleura and recesses

7. Mediastinum–Superior, Anterior, posterior, middle Mediastinum, Trachea, Larynx, Principal Bronchus

8. Lungs-: Lobes and Fissures–Right Lung, Left lung

9. Bronchopulmonary Segments-Blood Supply to the lungs, Lymphatic drainage of the lungs, Nerve supply of the lungs

10. Nerve supply of Abdomen and pelvis- Lumbar & sacral plexus in detail.

11. Lymphatics of Abdomen and pelvis

12. Peritoneal Cavity– Peritoneum, Folds and Recesses

13. Urinary system: Organs –Kidney, Ureter, Bladder and Urethra

14. Male Reproductive system

15. Female Reproductive system

16. Regional–Cubital Fossa, Inguinal canal, Femoral Triangle and Adductor canal

17. Embryology: Urogenital System-Development of-Kidney, ureter, urinary bladder, urethra, prostate, uterus, uterine tube, vagina, male external genitalia, female external genitalia

18. Histology of kidney, ureter, urinary bladder & urethra

PHYSIOLOGY (60 Hours)

Introduction to physiology: The cell and general physiology-Functional organization of the human body and control of the "Internal Environment"-Homeostasis-Regulation of body functions the cell and its function-Physical structure of the cell-Functional systems of the cell Genetic control of protein synthesis, cell function and cell reproduction-RNA-Cell differentiation

Temperature regulation: Body Temperature-Heat production & loss- role of hypothalamus-neuronal effectors mechanism-body temp-Behavior control-local skin temperature-Fever-Exposure to cold.

Nephro-Endocrinology Introduction to endocrinology The pituitary hormones and their control by the hypothalamus: The pituitary Gland and its relation to the hypothalamus – control of pituitary secretion by the hypothalamus – physiological functions of growth hormone – The posterior pituitary gland and its relation to the hypothalamus The thyroid metabolic hormones: Formation and secretion of the thyroid hormones – functions of the thyroid hormones in the tissues – regulation of thyroid hormone secretion – disease of the thyroid. The adrenocortical Hormones: Chemistry of Adrenocortical secretion – Functions of the mineralocorticoids-aldosterone – Functions of the glucocorticoids – The adrenal androgens – Abnormalities of adrenocortical secretion. Insulin, Glucagon and diabetes mellitus: Insulin and its metabolic effects Glucagon and its functions – somatostatin – hyperinsulinism. Parathyroid hormone, Calcitonin, calcium and phosphate metabolism, vitamin D, Bone and Teeth: calcium and phosphate in the extracellular fluid and plasma-function of vit D – Bone and its relationships to extracellular calcium and phosphates – parathyroid hormone – calcitonin – overall control of calcium Ion concentration – physiology of parathyroid and bone diseases

The Kidneys and Body Fluids

The body fluid compartments: extracellular and intracellular fluids interstitial fluid and Edema

a. Body fluid compartments

b. Constituents of extracellular and intracellular fluids

c. Osmotic equilibrium and fluid shifts between the extracellular and intracellular fluid

d. Changes in the volumes and osmolalities of the extracellular and Intracellular fluid compartments in abnormal states

e. Edema fluids in the potential spaces of the body

Formation of urine by the kidney: Renal Blood Flow, Glomerular filtration and their Control.

a. Physiologic anatomy of the kidney

b. Basic theory of nephron function

c. Renal blood flow and pressures

d. Glomerular filtration and the glomerular filtrate

e. Control of the glomerular filtration rate and renal blood flow

f. Reabsorption of fluid by the peritubular capillaries

Formation of urine by the kidney: Processing of the filtrate in the tubules

a. Effect of tubular load and tubular transport maximum on urine constituents

b. The concept of Plasma Clearance its use in assessing renal function

Renal associated mechanism for controlling extracellular fluid osmolality and sodium concentration.

a. The mechanism for excreting excess water: Excretion of a dilute urine

b. The mechanism for excreting excess solutes: The countercurrent mechanism for excreting a concentrated urine

c. Control of extracellular fluid osmolality and sodium concentration

d. Sodium excretion and its control by aldosterone

Renal regulation of Blood volume and extracellular fluid Volume: Excretion and regulation of urea, potassium, and other substances

a. control of blood volume

b. control of extra cellular fluid volume

c. urea excretion

d. Potassium excretion

e. Control of the extracellular concentrations of other ions

Regulation of Acid-Base Balance

a. Function of Acid – Base Buffers

b. Respiratory regulation of Acid – Base balance

c. Renal control of Hydrogen Ion concentration

d. Clinical abnormalities of Acid-Base Balance

Renal Disease, Diuresis, and Micturition

a. Renal Disease

b. Renal Function tests

c. Diuretics and mechanisms of their actions

d. Micturition

Biochemistry (60 Hours)

1. Chemistry and metabolism of Carbohydrates
2. Chemistry and metabolism of Lipids
3. Chemistry and metabolism of proteins
4. Chemistry and metabolism of nucleic acids
5. Enzymes and vitamins, biological oxidation
6. Water and minerals
7. Hemoglobin chemistry and metabolism
8. Acid- Base balance
9. Renal functions tests(Renal function Formation and composition of urine–Specific gravity and pH, solids and 24 hour urine, abnormal constituents of urine- Glycosuria-glycosuria, fructosuria, pentosuria, lactosuria, galactosuria. Proteinuria, Ketone bodies, Bile pigments and bile salts, Blood, porphyrins, Urinary lithiasis).
10. LFT
11. Detoxification
12. Hormones
13. Energy metabolism
14. Elements of Biochemical Genetics & Inborn errors of metabolism
15. Classification, mechanism of action of hormones regulating hormones calcium and phosphorous.
16. Regulation of pH of Blood and body fluids: Regulatory mechanisms, Renal Mechanism, Disturbances in acid-base balance, respiratory acidosis, respiratory alkalosis, metabolic acidosis, metabolic alkalosis, assessment of the acid-base balance, carbon-dioxide combining power of blood, alkali reserve, anion gap
17. Overview of estimation of serum creatinine, blood urea, serum electrolytes, 24 hrs. Urinary protein –Lab methods.

Microbiology (60 Hours)

Objectives:

1. To have the basic knowledge about the general characteristics of bacteria.
2. To understand the principles of immunology.
3. To learn about the application of Microbiology and Immunology in dialysis.

Theory

1. Classification, Morphology, and nutritional requirements of bacteria.

2. Sterilization, disinfection, and antimicrobial agents.

3. Normal flora of human body, infection, sources, and modes of transmission of infection.

4. Immunity: Innate immunity, acquired immunity, humoral and cell mediated immunity.

5. Antigen, antibody & complement.

6. Structure and functions of immune system and immune response.

7. Hypersensitivity and transplant immunology.

8. Pyogenic cocci.

9. Aerobic and anaerobic spore bearers & Corynebacterium.

10. Mycobacteria.

11. Enterobacteriaceae, Pseudomonas & Vibrio

12. Spirochetes.

13. General properties of viruses.

14. Hepatitis, HIV & CMV.

15. Oncogenic viruses & antiviral drugs.

16. Opportunistic, parasitic, and fungal infections and Nosocomial infections.

17. Diagnostic microbiology: An approach to lab diagnosis of microbial infections.

18. Hospital infection- Causative agents, transmission methods, investigation, prevention, and control

19. Biomedical waste management

Renal genetics and molecular biology (60 Hours)

1. Introduction, Cell, Cell Organelles, cell Cycle, and Cell division
2. Nucleic Acids – Structure, Function, and Chemical Properties, DNA, RNA
3. Introduction to Clinical Genetics, Glossaries, Genome Organization, Mutations:- Types, Pedigree analysis, Inheritance pattern, Genetic syndromes, and Inborn errors of metabolism
4. Genetics of Renal disorders, polycystic kidney, Hereditary nephropathy, Urinary tract malformation, Renal Stones, Renal transport disorder, and concept of Counseling
5. Introduction to Molecular Biology

a. Concept

b. Molecular Biology

6. Identification of the genetic materials
7. Chemical nature of genetic materials
8. Replication of DNA
9. Non-genetic ribonucleic Acid(RNA)and transcription
10. Genetic code
11. Protein synthesis
12. Regulation of gene action
13. Stem cell research: principles and application

Anatomy Practical (60 Hours)

Spotters:

a. Identification of arteries.
b. Identification of veins.
c. Identification of organs.
d. Identification of muscles of abdomen.
e. Identification of vertebra, ribs and pelvis.

Discussion:

a. Development of urogenital system.
b. Histology of kidney, ureter, bladder and urethra.
c. Male and female pelvis.

Physiology Practical (60 Hours)

Hematology

1. Study of a compound microscope.
2. Estimation of hemoglobin (Shali‟s Method)
3. Determination of Total erythrocyte (RBC) count
4. Determination of RBC indices or blood standards
5. Determination of total Leucocytes (WBC) count
6. Preparation of peripheral blood smear and determination of different leucocyte count
7. Determination of BT & CT

8. Determination of Blood group (A, B, C, & Rh system)
9. Determination of specific gravity of blood 10. Determination of osmotic fragility of RBC
10. Determination of ESR & PCV
11. Determination of platelet count, reticulocyte, absolute eosinophil count Clinical examination

1. An introduction to clinical examination
2. Clinical examination of the abdomen
3. Clinical examination of the Respiratory system
4. Clinical examination of the CVS
5. Clinical examination of the NS

Cardio-Vascular System

1. Recording system arterial Blood Pressure
2. Effect of posture and exercise on Blood pressure
3. Measurement of blood flow by venous occlusion plethysmography

Biochemistry Practical (60 Hours)

1. Analysis of abnormal Urine
2. Estimation of urea, creatinine, uric acid, Sugar. Cholesterol
3. Demonstration – Agarose gel electrophoresis of Serum Protein, Arterial blood gas analysis

Microbiology Practical's (60 Hours)

1. Compound microscope.
2. Demonstration of sterilization equipment‟s: hot air oven, autoclave, bacterial filters.
3. Demonstration of commonly used culture media, nutrient broth, nutrient agar, blood agar, chocolate agar, MacConkey medium, L J media, Robertson cooked meat media, Potassium tellurite media with growth, MacConkey medium with LF & NLF, NA with staph.
4. Anaerobic culture methods.
5. Antibiotic susceptibility test
6. Demonstration of common serological tests: Widal, VDRL, ELISA.
7. Gram staining.
8. Acid fast staining.
9. Fungal Staining
10. Stool exam for helminthic ova & cysts.

11. Visit to hospital for demonstration of work done by infection control team and biomedical waste management department.

Residency –I (90 Hours)

In the residency the professional is expected to work and contribute to the Dialysis Therapy unit.

Areas to be Covered.

1. Blood Bank
2. PHC, CHC
3. OPD
4. Transplant Work Up
5. Emergency Medicine
6. Radiology
7. Psychiatric
8. Operation Theater for assisting in AVF, AVG Creation, Renal Transplantation and PD Catheterization

Second Semester

Renal Pathology (60 Hours)

1. Gross appearance, Histology of Kidney, outline and functions of kidney

2. Names of congenital anomalies and acquired cystic diseases

3. Broad outline of pathogenesis of Glomerular diseases. Immune mechanism- Types

4. Classification of glomerulonephritis – Names of the types with mention of pathogenesis

5. Clinical manifestation of glomerulonephritis

6. Glomerular lesions in systemic disease. List of disease and brief outline of lesions

7. Diseases affecting tubules and interstitium. Acute tubular necrosis etiology and clinical manifestations

8. Pyelonephritis and urinary tract infection. Etiology and clinical manifestations

9. Chronic pyelonephritis, reflux nephropathy. Etiopathogenesis and clinical manifestations

10. Drug and toxin induced nephropathy. Etiopathogenesis and clinical manifestation

11. Benign and malignant nephrosclerosis. Etiopathogenesis, clinical manifestation

12. Urolithiasis – Types of stones, pathogenesis

13. Obstructive uropathy. Etiopathogenesis clinical features

14. Acute renal failure

15. Chronic renal failure

16. Tumors of the kidney and urinary bladder classification (Names)

17. Urine analysis – Introduction

18. Physical

19. Chemical qualitative and quantitative

20. Urine analysis, Microscopic

21. Peripheral blood smear

Pharmacology (60 Hours)

General Pharmacological principle

1. Definitions, routes of drug administration
2. Pharmacokinetics
3. Pharmacodynamics

4. Adverse drug effects

Drugs acting on autonomic nervous system

1. ANS – general considerations

2. Cholinergic system and drugs

3. Anticholinergic drugs

4. Adrenergic system

5. Antiadrenergic Drugs

Autacoids

1. Analgesics

2. Opioid

3. Nonopioid

Bronchodilators

Hormones

Steroids

Central Nervous system

1. Local Anesthetics

2. General anesthetics

3. Ethyl and methyl alcohols

4. Sedative-hypnotics

5. Antiepileptic drugs

6. Drugs used in Mental illness

Cardiovascular drugs

1. Drugs affecting Renin- Angiotensin system and plasma kinins

2. Cardiac glycosides and drugs for CHF

3. Antiarrhythmic drugs

4. Antianginal drugs

5. Antihypertensive drugs

6. Diuretics

Blood

1. Hematinic and erythropoietin

2. Drugs affecting coagulation, bleeding and thrombosis
3. Hypolipidemic drugs and plasma expanders

Gastrointestinal drugs

1. Drugsforconstipation and diarrhea

Chemotherapy

1. Antimicrobial drugs: General consideration
2. Sulfonamides, clotrimazoles and quinolones
3. Beta lactam antibiotics
4. Tetracycline and chloramphenicol
5. Aminoglycoside antibiotics
6. Macrolide and other antibacterial antibiotics treatment of UTI
7. Antitubercular drugs
8. Antifungal drugs
9. Antiviral drugs
10. Anticancer drugs
11. Immuno suppressants, gene therapy,
12. Chelating agents
13. Vaccines

Nephro-immunology (45 Hours)

Nature of the Immune system

1. Historical concepts and introduction to serological testing:

b.Immunity and immunization

a. Cellular verses humoral immunity
b. Antigens and haptens
c. Cells – mediated immunity

C.The Age of serology

d.Other historical developments

2. Natural Immunity:

a.External defense system

b.Internal defense system

- Cellular Defense mechanism
- Acute phase reactants
- Inflammation

c.Summary

3. The lymphoid system:

a.Primary lymphoid organs

- Bone marrow
- Bursa of Fabricius
- Thymus

b.Secondary lymphoid organ

- Spleen
- Lymph nodes
- Other secondary organs

c.Surface marker on lymphocytes

d.Stages in B-cell differentiation

- Pro-B cells
- Pre– B cells
- Immature B cells

- Mature B cells
- Activated B cells
- Plasma cells

E. T cells Differentiation

- Double– negative stage
- Double positive stage
- Mature T cells
- Antigen activation

f. Third population or natural killer

- Mechanism of cytotoxicity
- Antibody– Dependent cell

g. Laboratory identification of lymphocytes

- Fluorescence Microscopy
- Cell flow cytometry
- Other methods

h.summary

4. Nature of Antigens and the MHC Complex

a. Factors influencing the immune response

b. Traits of Immunogens

c. Nature of epitopes

d. Haptens

e. Relationship of antigens to the Host

f. Adjuvants

g. Major Histocompatibility Complex

- Genes coding for the HLA antigens
- Class I Antigens
- Class II antigens
- Antigen presentation
- MHC and Autoimmunity

5.Antibody structure and Function:

a. Tetra peptide structure of immunoglobulin
    - Cleavage with papain
    - Pepsin digestion
b. The Nature of Light Chains
c. Heavy chain sequencing
d. Antigen recognition unit
e. Hings region
f. IgG
g. IgM
h. IgA
i. IgD
j. Ige
k. Theories of antibody diversity
l. Genes coding for immunoglobin
m. Monoclonal antibody
n. Summary

6.Cell-Mediated Immunity:

A. Activation of T Helper cells
- Adhesion Molecules
- Specific Antigen recognition
- Function of Interleukin-2
- Triggering of Interleukin-1
- T- Helper subsets

B.Stimulation of B cells

C.Activation of cytotoxic T cells
- Role of cytokines in the inflammatory response
- I-1, I-2, I-3, I-4, I-5, I-6, Other I , Interferon‟s, TNF, other factor

D.Laboratory determination of T lymphocyte function

E. Summary

7. Complement:

A.The classic pathway

- The recognition unit
- The activation unit
- The membrane attacks

B.The alternative pathway

C.System controls

- fluid phase regulators
- cell – bound regulators

D.Other cell membrane Receptors

E.Biologic manifestation of complement activation

F.Complement and Diseases states

G.Complement Deficiencies

- Paroxysmal Nocturnal Hemoglobinuria
- Other complement

H.Laboratory detection of complement abnormalities

I.Summary

8.Transplantation:

A.Review of MHC

B.Transplantation

- Descriptive terms
- Graft rejection
- Tissue matching
- Types of tissues and organs transplanted
- Prevention and treatment of rejections
- Handson training in HLA typing

Research methodology and biostatistics (60 Hours)

Introduction

a. Meaning of research, Objectives of research, Types of research, Significance of research, Research Process, Criteria of research

Defining research problem

a. Selection of the problem, Necessity of defining the problem, Techniques in selecting a problem, Examples.

Research Design

a. Meaning of design, Need for a research design, Different research designs, Basics of experimental designs, Developing research plan.

Sampling Design

a. Sample and Universe, Sampling, Different methods of sampling, Selection of a random sample, Census and survey.

Measurement and scaling techniques

a. Measurement in research, Measurement tools, Scaling and scaling techniques.

Data collection

a. Methods of data collection, Questionnaire/schedule, Selection of appropriate method for data collection, Guidelines for constructing questionnaire, Guidelines for interviewing, Sample questionnaires

Data Analysis

Introduction to data analysis, Demo with SPSS software. Dissertation/Reporting (report writing)

Technique of interpretation, Report writing/Dissertation writing, Steps in report writing, How to critique an article.

Biostatistics

1. Statistics: What is statistics – Importance of statistics in behavioral sciences – Descriptive statistics and inferential statistics – Usefulness of quantification in behavioral sciences.
2. Measurements – Scales of measurements – Nominal, Ordinal, Interval and Ratio scales.
3. Data collection – Classification of data – Class intervals – Continuous and discrete measurements – Drawing frequency polygon – types of frequency polygon – Histogram – Cumulative frequency curve - Ogives – Drawing inference from graph.
4. Measures of central tendency – Need – types: Mean, Median, Mode – Working out these measures with illustrations.
5. Measures of variability – Need – Types: Range, Quartile deviation, Average deviation, Standard deviation, Variance – Interpretation.
6. Normal distribution – General properties of normal distribution – Theory of probability – illustration of normal distribution – area under the normal probability curve.
7. Variants from the normal distribution – skewness – Quantitative measurement of skewness – kurtosis – measurement of kurtosis – factors contributing for non-normal distribution.

8. Correlation – historical contribution – meaning of correlation – types: - Rank correlation, Regression analysis.
9. Tests of significance- need for – significance of the mean – sampling error- significance of differences between means – interpretation of probability levels – small samples – large samples - Inferential statistics – Parametric & Non parametric methods – Elements of multivariate analysis

Biophysics and biomedical instrumentation (30 Hours) Part A- Biophysics

1. Properties of matter

2. Membranes

3. Cells, structure of cells, structure of membranes, membrane transport processes, electrical properties of membranes - Structure & Properties of Ion Channels Voltage, Patch Clamp Tech – Synaptic Transmission- NMJ – Signal Transduction – Signaling – Second Messenger – Neural Network–Ion Channels in Brain Pathology.

4. Biophysics of nerve transmission

5. Membrane potential and its origin, membrane conductance, resistance and capacitance, action potentials, ionic basis of resting membrane potential and action potentials, synaptic transmission, electrophysiological basis for E.E.G., E.C.G & E.M.G recordings, evoked potential sand conduction velocity

6. Biophysics of the cardiovascular system

7. Basic principles of fluid dynamics, hemodynamics, introduction to principles of electrocardiography & vector cardiography

8. Biophysics of respiration

9. Mechanics of breathing, lung compliance, airway resistance and their measurement, surface tension& surfactants, partial pressure of gases, principles of gas exchange, measurement of blood gases, pulmonary function tests.

10. Biophysics of renal function
11. Diffusion, filtration, physical basis for ion, solute and water movement, dialysis, renal function tests.
12. Radiation biophysics
13. Introduction to nuclear physics and radioactivity, biological effects of radiation, radiation detection and measurement, radioisotopes and their uses.
14. Biophysical techniques
15. Introduction to microscopic techniques, introduction to spectroscopic techniques, introduction to imaging techniques, introduction to techniques in biophysical chemistry.

Part B- Biomedical instrumentation

1. Introduction to Biomedical Instrumentation

a. The Age of Biomedical Engineering

b. Development of BM instrumentation

c. Biometrics

d. Introduction to the man-instrument system

e. Components of the man-instrument system

f. Physiological systems of the body

g. Problems encountered in measuring a living system.

h. Some conclusions

i. The objectives of this book

2. Basic Transducer principles:

a. The transducer and transduction principles

b. Active transducers

c. Passive transducers

d. Transducers for Biomedical applications

3. Sources of Bioelectric potentials:

a. Resting and action potentials

b. Propagation of Action potentials

c. The Bioelectric potentials

4. Electrodes

a. Electrode theory

b. Biopotential electrodes

c. Biochemical transducers

5.. The computer in Biomedical Instrumentation:

d. The digital computer

e. Computer Hardware

f. Computer software

Microprocessors

a.Types of microprocessors

b.Microprocessors in Biomedical Instrumentation

- Calibration
- Table lookup
- Averaging
- Formatting and printout

Interfacing the computer and medical instrumentation and other equipment.

a. Digital interfacing requirement

b. Analog-to-digital and Digital-to-Analog conversion

Biomedical computer application

a.Data acquisition.

b.Storage and retrieval

c.Data reduction and transformation

d.Mathematical operation

e.Pattern recognition

f.Limit detection

g.Statistical analysis of data

h.Data presentation

i.Control function

- Computer analysis of the ECG
- The digital computer in the clinical chemistry laboratory.
- Digital computerized in hemodialysis machine
- Other computer application

6.Electrical safety of Medical Equipment.

a. Physiological effects of electrical current.

b. Shock hazards from electrical equipment.

c. Methods of accident prevention

- Grounding
- Double insulation
- Protection by low voltage
- Ground – fault circuit interrupter

- Isolation of patient – connected parts
- Isolated power distribution systems

7. Patient care and monitoring:

a. The elements of intensive – care monitoring - Patient monitoring displays

b. Diagnosis, calibration and repairability of patient – monitoring equipment

c. Other instrumentation for monitoring patients

d. The organization of the hospital for patient care monitoring

e. Defibrillator

Table of Contents

1. Description of Machine Functions and Malfunction

a. Description of the T1Test

b. Function description of the modules

c. Function description of the hydraulic unit

2. Technical safety checks and maintenance

a. General notes

b. Technical safety checks and maintenance procedures

c. VDE test 4008E, 4008H

d. VDE test 4008B,4008S

e. TSC and maintenance checklist

3. Adjustment

a. OVERVIEW of the Dip switches in the 4008E

b. Calibration mode

c. Hydraulics

d. Air detector

4. Calibration Program

5. Diagnostics Program

a. General notes

b. Menu structure

c. Reading the analog inputs of CPU

d. Reading the digital inputs of CPU

e. Writing the analog outputs of CPU

f. Writing the digital outputs of CPU

g. Writing/ Reading the digital output of CPU

6.Setup Menu

a.Overview

b.Main menu

7.Miscellaneous

8.Circuit diagram and circuit description

a. Block diagram

b. Ac diagram 4008E/H

c. AC diagram 4008B/S

d. Block diagram of voltage supply Block diagram of screen 4008H/S

e. Connection layout diagram

9.P.C.B. LP

10.BASICS

1. Basics in Electricity

- Charge
- AC & DC
- Voltage, Current, Resistance, Capacitance & Inductance
- Electrical Shock hazard including Leakage currents
- Voltage stabilizer and Uninterrupted power supply

2. Basics in Electronics

- Semiconductions
- Rectification, Amplification, Oscillation & switching
- Digital Electronics
- Microprocessor & micro controllers
- Memory [RAM, ROM, EPROM]

3.Operating microscopy, Lab microscopy, Electron microscope

4.General instrumentation for OT

Biomedical instrumentation Practical (30 Hours)

1. AC, DC
2. Measurement of Resistance, capacitance, Inductance
3. Measurement of Leakage of current
4. Amplification, Solid state, Digital
5. Trouble shooting equipment failure

Basic course in entrepreneurship (45 Hours) (Elective Course)

LEARNING OBJECTIVES:

- To understand the fit between you and your entrepreneurial ambitions
- To find a problem worth solving
- To identify your customers
- To develop a solution for your customers' problems and problem solution
- To build and demonstrate an MVP
- To structure a business model around the problem, customer, and solution and present your Business Model Canvas

1. ORIENTATION

What is entrepreneurship – myths about entrepreneurship – impact of an entrepreneur and social entrepreneurship – wealth building and making an impact

IDEA/PROBLEM

What is a business opportunity and how to identify it - Methods for finding and understanding problems - (Observation, Questioning, DT, Jobs to be done (JTBD) - Introduction to Design Thinking - Process and Examples - Generate ideas that are potential solutions to the problem identified

2. CUSTOMER

The difference between a consumer and a customer (decision maker); Market Types, Segmentation and Targeting, Defining the personas; Understanding Early Adopters and Customer Adoption Patterns - Identify the innovators and early adopters for start-up - Basics of Lean Approach and Canvas; Types of Business Models (b2b; b2c)

3. BUSINESS MODEL AND VALIDATION

Introduction to Risks; Identify and document your assumptions (Hypotheses); Identify the riskiest parts of Plan - Develop the Solution Demo - Sizing the Opportunity - Building an MVP (Minimum Viable Product)

4. MONEY AND TEAM

Revenue Streams: Basics of how companies make money - Understand income, costs, gross and net margins - Identify primary and secondary revenue streams - Pricing and Costs - Financing Your New Venture - Team Building: Role of a good team in a venture's success; What to look for in a team; How do you ensure there is a good fit? Defining clear roles and responsibilities

5. MARKETING AND SALES

Positioning – channels and strategy – sales planning – Importance of project management to launch and track progress - Understanding time management, workflow, and delegation of tasks– Business regulation: Basics of business regulations of starting and operating a business

- Importance of being compliant and keeping proper documentation.

BASICS OF HOSPITAL MANAGEMENT (45 Hours) (Elective Course)

1. Introduction to hospitals, functions, and managerial aspects of Hospitals

   Evolution of hospitals and the concept of modern hospital

   a. Explain how hospitals originated and transformed itself throughout history.
   b. What is the role of hospitals in healthcare
   c. Explain health delivery system in India
   d. Explain the types of hospitals and how modern hospitals are planned

2. General functions of hospital store management

   Material management and Inventory control

   a. Define basic concepts of material management.
   b. List the functions of hospital stores
   c. Explain the types of indents during the procurement process
   d. Explain various inventory management techniques

3. Analysis of Hospital Organization

1. Administration of Clinical and nonclinical services

   a. Explain OPD and Inpatient services
   b. Interpret on hospital pharmacy services
   c. Explain Emergency Services
   d. Outline the services of Transport department in the hospital
   e. Explain Intensive Care Unit (ICU)
   f. Outlines the services of Operation Theater in the hospital
   g. Explain the various Radiological Services in the hospital

h. Explain the Dialysis services in a hospital

i. Explain biomedical waste management in hospital

j. Explain the functions of CSSD & Medical gas supply

k. What are the various Committees in hospital

l. Outline the services of blood bank in a hospital

m. Explain the Disaster management in a hospital

4. Public relation and Laws in Hospital administration

Public relation and Laws

a. Explain different laws in hospital administration like, COPRA, PCPNDT ACT, POCSO ACT and RTI etc

b. Explain the roles and responsibilities of public relations in hospital

c. List out the various risk involved in patient care

d. Explain the elements of disaster management

e. Plan for a Mock drill during a disaster strike in a hospital

5. Managerial Economics

Cost-benefit analysis, Cost effectiveness analysis, cost utility analysis

a. Explain cost analysis

b. Explain cost effectiveness

c. Explain cost utility

Break even analysis

a. Break even analysis of CT scan

b. Break even analysis of dialysis machine

6. NABH Accreditation

a. Explain NABH accreditation as quality management

7. Strategic Management Tools

a. Explain strategic management tools

8. Staffing Principles

a. Explain Staffing

Renal Pathology Practical's (30 Hours)

1. Gross appearance, Histology of Kidney, outline and functions of kidney congenital anomalies and acquired cystic diseases
2. pathogenesis of Glomerular diseases. Immune mechanism- Types
3. Clinical manifestation of glomerulonephritis
4. Glomerular lesions in systemic disease
5. Pyelonephritis and urinary tract infection
6. Drug and toxin induced nephropathy
7. Benign and malignant nephrosclerosis
8. Urolithiasis
9. Obstructive uropathy.
10. Acute renal failure
11. Chronic renal failure
12. Tumors of the kidney and urinary bladder
13. Urine analysis
14. Peripheral blood smear

Pharmacology Practical's (30 Hours)

1. Administration of Erythropoiesis stimulating agents (ESA)
2. IV Iron Preparation
3. Vitamin supplements
4. Crystalloids & Colloids
5. Inotropic drugs
6. Anti-pyretic drugs
7. Anti-emetic drugs
8. Heparin, LMWH and Tri-sodium citrate, Regional Anticoagulation
9. Chemicals used in dialysis unit
10. PD fluids
11. Dialysis & CRRT solutions
12. Emergency Drugs used in Dialysis

Nephro-immunology Practical's (30 Hours)

Nature of the Immune system

1. serological testing
2. Immunity and immunization
3. The lymphoid system:
4. Laboratory identification of lymphocytes
    - Fluorescence Microscopy
    - Cell flow cytometry
    - Other methods
5. Genes coding for the HLA antigens
6. Laboratory determination of T lymphocyte function
7. Laboratory detection of complement abnormalities
8. Handson training in HLA typing

Residency –II (135 Hours)

In the residency the professional is expected to work and contribute to the Dialysis Therapy unit. Areas to be Covered.

1. Blood Bank

2. PHC, CHC

3. OPD

4. Transplant Work Up

5. Emergency Medicine

6. Radiology

7. Psychiatric

8. Operation Theater for assisting in AVF, AVG Creation, Renal Transplantation and PD Catheterization

Third Semester

Renal Disease (60 Hours)

1. Patient Assessment
    a. Physical Diagnosis
    b. Urinalysis/ BUN, Creatinine

c. Measurement of Glomerular Filtration Rate
d. Measurement of Urinary Protein
e. Renal Imaging Techniques
f. Renal Biopsy
g. Indications for Dialysis
h. Drug Therapy in Renal Disease

2. Clinical Syndromes
a. Acute Glomerulonephritis/Acute interstitial nephritis
b. Nephrotic syndrome
c. Etiology, Pathophysiology, and Diagnosis of Acute Renal Failure
d. Management of Acute Renal Failure
e. Obstructive Uropathy
f. Asymptomatic Proteinuria
g. Asymptomatic Hematuria
h. Rapidly Progressive Glomerulonephritis
i. Nephrolithiasis
j. Urinary Tract Infection
k. Disorders of Tubular Function

3. Primary Glomerular Disease
a. Minimal Change Disease
b. Focal Segmental Glomerulosclerosis
c. Membranous glomerulopathy
d. IgA Nephropathy
e. Membranoproliferative Glomerulonephritis

4. Secondary Glomerulonephritis
a. Diabetic Nephropathy
b. Lupus nephritis
c. Post infectious glomerulonephritis
d. Hepatitis-Associated Glomerulonephritis

e. HIV-Associated Renal Disorders

5. Other parenchymal renal diseases
    a. Renal Dysplasia
    b. Cystic Diseases of the Kidneys
    c. Other hereditary renal diseases
    d. Reflux Nephropathy
    e. Renal Vasculitis
    f. Other Vascular Renal Disorders
    g. Sickle Cell Nephropathy
    h. Renal Disease due to dysproteinemias
6. Renal Diseases in Pregnancy
    a. Hypertension-Essential
    b. PIH/Eclampsia
    c. HELLP Syndrome
    d. ARF in pregnancy
    e. Lupus in Pregnancy
    f. Postpartum HUS
7. Toxic Nephropathy
    a. Drugs: Aminoglycosides/NSAIDS/Antineoplastic drugs
    b. Radiocontrast
    c. Heavy metals- Lead, Mercury, Cadmium.
    d. Alternative medicine
    e. Snakebite–venom
    f. Drug overdose- Barbiturates, Diazepam, Beta Blockers
8. Tropical Diseases related to Kidney
    a. Malaria
    b. Filariasis
    c. HANSENS Diseases
    d. Leptospirosis
    e. Dengue

f. Leishmaniasis

9. Geriatric Kidney Diseases
    a. Aged kidney
    b. Hypertension
    c. Atherosclerotic Kidney Diseases
10. Chronic Kidney Diseases
    a. KIDOQI Guidelines of definition, grading and management
    b. Epidemiology and outcomes of End-Stage Renal Diseases
    c. Uremic toxins
    d. Pathophysiology of Chronic Kidney Diseases
    e. Renal Osteodystrophy
    f. Renal Anemia
    g. Other manifestations of Uremia
    h. Malnutrition
    i. Risk factors– (Traditional and non-traditional) For Cardiovascular mortality in ESRD patients
11. Hypertension
    a. Essential Hypertension
    b. Renal parenchymal hypertension
    c. Renovascular Hypertension
    d. Pheochromocytoma
    e. Other Causes of Secondary Hypertension
    f. Hypertensive Emergencies
    g. Childhood Hypertension
12. Acid-Base and Electrolyte Disorders
    a. Metabolic Acidosis
    b. Metabolic Alkalosis
    c. Respiratory Acidosis
    d. Respiratory Alkalosis
    e. Hyponatremia and Hypernatremia
    f. Hypokalemia and Hyperkalemia

g. Hypocalcemia and Hypercalcemia

h. Phosphorus

i. Magnesium

13. Approach to Patient with Renal Failure Manifestations of renal failure:

a. Systemic consequence of renal failure Alterations in following including signs, symptoms etiology &management.

b. Uremia

c. Fluid balance

d. Electrolyte and ion balance (sodium, potassium, calcium, phosphate, Magnesium, hydrogen, bicarbonate, aluminum)

e. Cardiovascular system (Hyperkalemia, Hypertension, Pericarditis, Pericardial effusion, Pericardial tamponade)

f. Respiratory system

g. Gastrointestinal system

h. Hematological system (Anemia, Platelet dysfunction)

i. Endocrine function

j. Neuromuscular function & sleep problems

k. Skeletal system

l. Psycho-social function

m. Dermatological manifestations

14. End-Stage Renal Disease Management

a. Technical Aspects of Hemodialysis

b. Hemodialysis: Assessing Adequacy

c. Complications of Hemodialysis

d. Technical Aspects of Peritoneal Dialysis

e. Complications of Peritoneal Dialysis

15. Contraindications for Dialysis (Including PD)

a. Alzheimer‟s Disease

b. Multi-infract denaturation.

c. Hepatorenal syndrome

d. Advanced cirrhosis with encephalopathy

e. Advanced malignancy etc.

Emergency medicine, ACLS and Renal Nutrition & Dietetics (45 Hours) BLS

1. BLS in perspective
    a. The need for medical interventions
    b. The ultimate Coronary Care Unit
    c. Emergency Cardiac Care
    d. The chain of Survival
    e. Role of the American Heart Association
2. Cardio Pulmonary Function and actions for survival
    a. The Cardiovascular and Respiratory system
    b. Action for survival
3. Risk factors and prudent Heart living
    a. Risk factors for Heart Attack
    b. Prudent Heart Living
    c. Summary: The role of Prevention
4. Adult BLS
    a. Citizen response to Cardio-pulmonary Emergency
    b. Indication for BLS
    c. The sequence of BLS; Assessment, EMS activations and the ABC of CPR
    d. CPR performed by one rescuer and two rescuers
    e. Foreign – Body airway obstruction Management
    f. CPR: The Human Dimension
    g. BLS Research Initiative
5. Special Resuscitation Situation
    a. Stroke
    b. Hypothermia
    c. Near – Drowning
    d. Cardiac arrest associated with Trauma
    e. Electric shock and lightning stroke
    f. Pregnancy

g. Asphyxiation
h. Special techniques and pitfalls and complication
i. Unique situation

6. Pediatric BLS
   a. Epidemiology
   b. Injury prevention
   c. Prehospital care
   d. The sequence of Pediatric BLS - the ABC of CPR
   e. Activation of the EMS system obstructive
   f. Foreign Body airway
   g. BLS in Trauma

7. Ethical and Legal considerations
   a. Values in Decision Making
   b. Instituting and Discontinuing CPR
   c. Legal mandates
   d. Conclusions

8. Safety during CPR Training and actual rescue
   a. Disease transmission during CPR Training
   b. Disease transmission during actual performance of CPR

9. Automated External Defibrillation
   a. Importance of Automated External Defibrillation
   b. Overview of Automated External Defibrillation
   c. Advantage and Disadvantage of Automated External Defibrillation
   d. Use of Automated External Defibrillation during Resuscitation attempts
   e. Automated External Defibrillation treatment algorithm
   f. Post resuscitation care
   g. Training
   h. Maintenance of Skills
   i. Medical control
   j. Quality assurance

ACLS

1. Cardiopulmonary Resuscitation and Advanced Cardiac Life Support
    a. Basic Life Support
    b. General Considerations of Advanced Cardiac Life Support: Arrhythmia recognition and defibrillation-ventilation and airway management-route of drug administration-IV fluids-diagnose and correct the underlying cause of the arrest-internal cardiac compression-initiation and discontinuation of resuscitation.
    c. Specific Arrest Sequences in Advanced Cardiac Life Support: VF and Pulseless VT – Systole-Bradycardia-Pulseless electrical activity (PEA)-Tachycardias
    d. Post resuscitation Management
    e. Common Medications Used in Advanced Cardiac Life Support: Epinephrine-Atropine Sulfate-Lidocaine-Procainamide hydrochloride- Bretylium tosylate-magnesium sulfate-adenosine-Diltiazem or verapamil-Isoproterenol-Sodium Bicarbonate-Calcium
2. Critical Care
    a. Respiratory Failure: General considerations-pathophysiology-Blood gas analysis
    b. Oxygen therapy: Nasal prongs-venturi masks-Nonrebreathing masks-A continuous positive airway pressure mask-Bilevel positive airway pressure
    c. Airway Management and Tracheal Intubation: Airway Management-Endotracheal Intubation-Surgical airways
    d. Mechanical Ventilation: Indications-Initiation of mechanical ventilation-Management of problems and complications-Weaning from mechanical ventilation-Drugs commonly used during endotracheal intubation and mechanical ventilation
    e. Shock: Resuscitative Principles-Individual shock states
    f. Hemodynamic Monitoring and Pulmonary Artery Catheterization: Indications-obtaining a pulmonary capillary wedge tracing-acceptance of PAOP readings-transmural pressure-Cardiac Output-Interpretation of hemodynamic readings
3. Cardiac Arrhythmias
    a. Recognition and Management: Clinical diagnosis of arrhythmias-Electrocardiographic data-Bradyarrhythmia‟s-premature complexes-Tachycardia-
    b. Antiarrhythmic Drug Therapy: General Principles-Antiarrhythmic agents
    c. Related Topics: Syncope-Electro-Cardioversion-Cardiac Pacing-Anti-tachycardia devices

Renal Nutrition and Dietetics

Part I: Foods and food groups

Cereals, Pulses, Fruits and Vegetables, Milk and Milk Products, Nits and oils seeds, non-Vegetarian foods, Beverages, Alcohol

Part II: Energy and nutrients

Carbohydrates, Fats, Proteins, Fat Soluble Vitamins, Water Soluble Vitamins, Minerals (Na, K, Fe, Ca, P, Mg, Mn, I, Cu, Co, Chloride, Fluoride) Dietary Fiber and Water.

Part III: Nutrition Management

1. NUTRITION THROUGH LIFE CYCLE

   Pregnancy and lactation, Pediatrics, Geriatrics, Athletes.

2. CLINICAL DIETETICS Hospital diets Nutritional management for the following:

   Under Nutrition (PEM), Over nutrition, Anemia, GI disorders, Hepatic, Renal, Cardiac/ HTN, Diabetes mellitus, Gout, Enteral Nutrition, Parenteral Nutrition.

Part IV: Nutritional Assessment

1. Anthropometry
   a. Methods, Interpretation, Limitation
   b. Growth and Body Composition (Fat mass and Free fat mass)
2. Biochemical Markers of Nutritional Status
   a. Physical Examination
   b. Medical History
3. Evaluation of Nutrient intake data
   a. Table of RDA
   b. Evaluation of Nutrient intake of individual
   c. Evaluation of Nutrient intake for a Population Introduction
4. Introduction
   a. Nutritional assessment system
   b. Methods used in nutritional assessment
   c. The design of nutritional assessment system
   d. Evaluation of nutritional assessment indices
      i. Reference distribution
      ii. Reference limits

iii. Cutoff points

5. Food consumption of Individual
    a. Methods
    b. New development in measuring food consumption
    c. Selecting an appropriate method
    d. Summary
6. Evaluation of nutrient intake data
    a. Tables of recommended nutrient intakes
    b. Evaluating Nutrient intakes of individuals
    c. Evaluating the nutrient intakes of population groups
    d. Probability approach to evaluating nutrient intakes
7. Anthropometric assessment
    a. Advantages and limitations of anthropometric assessment
    b. Sources of error in nutritional anthropometry
    c. Evaluation of anthropometric indices
8. Anthropometric assessment of growth
    a. Growth measurement
    b. Indices derived from growth measurements
9. Anthropometric assessment of body composition
    a. Assessment of body fat
    b. Assessment of fat-free mass
10. Clinical assessment
    a. Medical history
    b. Physical examination
11. Nutritional assessment of hospital patients
    a. Methods based on single indices
    b. Prognostic nutritional indices
    c. Hospital prognostic indices
    d. Cluster analysis

e. Subjective global assessment the prognostic value of nutritional assessment indicesFood consumption of Individual

f. Medical history

g. Physical examination

12. Nutritional assessment of hospital patients

a. Methods based on single indices

b. Prognostic nutritional indices

c. Hospital prognostic indices

d. Cluster analysis

e. Subjective global assessment the prognostic value of nutritional assessment indices

Nephro-Radiological and imaging sciences (30 Hours)

1. X-ray & Contrast Studies related to renal studies:
2. UROGRAPHY:Urographiccontrastmedia–choice–structure–contrastselection–contrastreaction rate of administered contrast material – methodology of intravenous urography – indications – contraindications – interpretation.
3. ULTRASOUND: Elementary ultrasound methodology–sonographic imaging characteristics – Doppler ultrasound – scanning techniques – ultrasound of the normal kidney – acute renal failure –surgical – medical renal disease – renal masses – cysts- solid renal masses – calculi – intra renal hematomas – renal transplant evaluation – rejection – renal infections – ultrasound versus intravenous contrast studies – percutaneous vs. guided ultra ventions – Doppler ultrasound in evaluation of renal vascular disease – evaluation of anatomic abnormalities.
4. RADIONUCLIDE RENOGRAPHY: Methodology – selection of radio nuclide – the gamma camera –indications – limitations – quantifications of renal function – GFR – effective renal plasma flow –obstructive uropathy–vesicoureteral reflux–renal transplantation–acute renal failure–Renovascular HT–captopril renography– static renal imaging.
5. COMPUTED TOMOGRAPHY OF THE KIDNEY: Anatomy – technique – renal masses – cystic masses – solid renal masses – tumors of the renal pelvis – renal calculi – obstructive uropathy –infarction–acute renal cortical necrosis–renal vein thrombosis–renal artery stenosis–acute pyelonephritis – renal and peri nephric abscess – emphy sematous pyelonephritis – pylo nephrosis –xanthogranulomatous pyelonephritis–tuberculous pyelonephritis–congenital anomalies–renal trauma–transplant kidneys
6. MAGNETIC RESONANCE IMAGING: technical aspects – magnetic resonance with IV contrast – normal kidney–congenital anomalies–obstruction–injection–renal parenchymal disease–hypertension and renal vascular disease – renal cysts – benign neoplasm‟s – malignancies of kidney –transplantation – magnetic resonance spectroscopy.

7. RENAL ANGIOGRAPHY: Indications– relative contraindications – patient preparation – basic procedure – specific techniques – abdominal aortography – selective renal arteriography –inferiorvenocavagraphy–selectiverenalvenography–renalveinrenninsampling– intraarterial–digital
8. Subtractionangiography–IV digital subtraction angiography–Complications–out patient arteriography– Angioplasty– renal artery stunts– transcatheter embolization.
9. RENAL BIOPSY: techniques of percutaneous renal biopsy – choice of needle – biopsy technique –trans jugular renal biopsy – open renal biopsy – contraindication – complication – processing of the biopsy specimen– malignancy of specimen interpretation.
10. NUCLEAR MEDICINE
    a. Basic of nuclear medicine.
    b. DTPA, DMSA.
    c. Peritoneal scintigraphy.
    d. Reflux scintigraphy.
    e. Transplant renal scintigraphy

Renal Disease Practical (30 hours)

1. Patient Assessment
2. Clinical Syndromes
3. Primary Glomerular Disease
4. Secondary Glomerulonephritis
5. Other parenchymal renal diseases
6. Renal Diseases in Pregnancy
7. Toxic Nephropathy
8. Tropical Diseases related to Kidney
9. Geriatric Kidney Diseases
10. Chronic Kidney Diseases
11. Hypertension
12. Acid-Base and Electrolyte Disorders
13. Approach to Patient with Renal Failure
14. End-Stage Renal Disease Management
15. Contraindications for Dialysis

Emergency Medicine and Renal Nutrition & Dietetics Practical's (30 Hours)

1. Adult BLS
2. Special Resuscitation Situation
3. Pediatric BLS
4. Automated External Defibrillation
5. Advanced Cardiac Life Support
6. Post resuscitation Management
7. Critical Care
8. Cardiac Arrhythmias
9. Nutrition Management
10. Nutritional Assessment

Nephro- Radiology and Imaging sciences Practical's (30 Hours)

1. X-ray & Contrast Studies related to renal
2. UROGRAPHY
3. ULTRASOUND
4. RADIONUCLIDE RENOGRAPHY
5. COMPUTED TOMOGRAPHY OF THE KIDNEY
6. MAGNETIC RESONANCE IMAGING
7. RENAL ANGIOGRAPHY
8. Subtractionangiography
9. RENAL BIOPSY
10. Peritoneal scintigraphy
11. Reflux scintigraphy.
12. Transplant renal scintigraphy

Residency part – III (225 Hours)

In the residency the professional is expected to work and contribute in the Nephrology OPD, Nephrology Ward, ICUs & Dialysis Therapy unit.

Fourth Semester

Advanced Renal Replacement Therapy (60 Hours)

1. Hemodialysis

a. Vascular Access:

i. History

ii. Epidemiology

iii. Arteriovenous Grafts

iv. Arteriovenous Fistula

v. Vascular Access Monitoring and surveillance

vi. Cuffed Venous Catheters

vii. Treatment of Vascular Access Dysfunction

viii. Pharmacologic Prevention of Vascular Access Failure

b. Artificial Physiology:

i.General Principles of Hemodialysis

1. Clearance
2. Hemodialysis Membranes: Effects and Characteristics

ii.Components of the Extracorporeal Circuit

1. Blood circuit
2. Dialysate Circuit
3. Variations in the Extracorporeal Circuit
4. On-line monitoring
5. Dialysate Composition
6. Water used in Hemodialysis

iii.The Dialysis Prescription

1. Dialyzer choice
2. Anticoagulation for Hemodialysis
3. Blood and Dialysate Flow
4. Dialysis Time
5. Dialysate composition
6. Dialysate Temperature

7. Ultrafiltration Rate

iv.Hemodialysis Adequacy

1. Historical Perspective
2. Low-Molecular-Weight Substances and Middle Molecules
3. The national cooperative dialysis study
4. Effect of Dialysis with High-Efficiency and High-Flux Dialysis
5. The Hemo Study

v.Alternative Chronic Hemodialysis Prescriptions

1. Daily Dialysis and Outcome
2. Nocturnal hemodialysis
3. Short daily hemodialysis
4. Outcome studies

vi.Alternative approaches to quantification of Dialysis

1. Urea reduction Ration and solute removal index
2. Standard measure of the Dose of Dialysis: Kt/V

vii.Management of the maintenance‟s hemodialysis patient

1. Uremic syndrome
2. Anemia
3. Cardiovascular disease
4. Vascular calcification
5. Calciphylaxis
6. Nutrition
7. Infection and immunity

viii.Complications of Hemodialysis

1. Hypotension
2. Cramps
3. Dialysis disequilibrium syndrome
4. Arrhythmias and angina
5. Hypoxia
6. Hypoglycemia

7. Hemorrhage
8. First-use syndrome or syndrome arrhythmias and angina
9. Problems with dialysate composition and integrity of the extracorporeal circuit

ix.Plasmapheresis

x.MARS

xi.The future of Renal Replacement Therapy

2.Peritoneal Dialysis

a. Components of the peritoneal dialysis system

i. Catheters
ii. Dialysis solutions
iii. Peritoneal membrane

b.Peritoneal Catheter Implantation

i. Surgical technique

ii. Bedside technique

c.Peritoneum as a dialysis system

i. Resistance to salt and water transport

ii. Models of peritoneal transport

iii. Physiology of peritoneal transport

d.Clinical observations of peritoneal membrane function

i. Characterization of peritoneal membrane transport

ii. Stability of the peritoneal membrane over time

iii. Conclusions

e.Clinical use of peritoneal Dialysis

i. Choice of Dialysis modality

ii. Peritoneal Dialysis Modalities

iii. Choice of peritoneal dialysis modality

f.Adequacy of peritoneal dialysis

i. When to initiate dialysis

ii. Measurement of the peritoneal dialysis dose

iii. Adequate dose of peritoneal dialysis

g.Writing the dialysis prescription

i. Initial prescription

ii. Adjustments to the initial dialysis prescription

iii. Further adjustments to the dialysis prescription

iv. Success in achieving adequate dialysis.

h.Nutrition

i. Malnutrition
ii. Nutritional parameters
iii. Nutrition guidelines
iv. Interventions for malnutrition

i.Peritoneal dialysis in special populations

i. Rapid transporters
ii. Elderly patients
iii. Patients with diabetes mellitus
iv. Black and Hispanic patients‟
v. Pediatric patients
vi. Patients infected with the human immunodeficiency Virus.
vii. Pregnant patients

j.Peritoneal dialysis in a Nonchronic setting.

i. Acute renal failure
ii. Nonnumeric uses of peritoneal dialysis

k.Complications of peritoneal dialysis

i. Cardiovascular disease
ii. Comparison of hemodialysis and peritoneal dialysis
iii. Indications for switching from peritoneal dialysis to hemodialysis.
iv. Noninfectious complications
v. Infections complications

l.Future trends in peritoneal dialysis

i. Mini PET

3.Clinical considerations in the Evaluation of Dialysis Patients

a. Hypertension in Dialysis Patients
b. Left Ventricular Dysfunction in Dialysis Subjects
c. Coronary Artery Disease in End-Stage Renal Disease
d. Autonomic Function and hemodynamic stability in End- Stage Renal Disease Patients
e. Infection and immunity in End-stage renal disease

f. B2-Microglobulin- Associated Amyloidosis of End-Stage Renal Disease

g. Renal Osteodystrophy

h. Dyslipidemias of End-Stage Renal Disease

i. Selection of Therapy for Patients with End-stage renal disease

j. Malnutrition and Intradialytic Parenteral Nutrition in ESRD Subjects

k. Disorders of Homeostasis in Dialysis Patients

l. Treatment of Anemia in Dialysis Subjects

m. Acquired Cystic Kidney Disease

n. Geriatric Dialysis Patients

o. Diabetic Dialysis Patients

p. Hemodialysis and Hemoperfusion for poisoning

q. Dialysis considerations in the patient with Acute Renal Failure

r. Infections in patients on Continuous Ambulatory Peritoneal Dialysis

s. Balancing outcomes in Dialysis with Economic Realities

4.Pediatric Dialysis

A.Pediatric HD

Introduction, HD in ARF causes of ARF in children, Indication for Dialysis, principles of Dialysis in ARF, HD in acute poisonings, HD for inborn errors of metabolism, acute vascular assess.

Chronic HD

incidence and etiology of terminal renal failure in pediatric population, patient selection, facilities for treatment, indication for dialysis, technical aspects of pediatric HD, vascular access, AV fistula, bridge grafts, catheters, complication of vascular access, HD equipment

Methods of Pediatric Dialysis: -

Common problem associated with long term HD, preparation of a child for eventual kidney Transplantation.

B. Pediatric PD

Peritoneal dialysis kinetics in children, catheter placement, PD in ARF, APD for reasons others

than ARF, PD in CRF, Intermittent PD, Continuous PD, CAPD, CCPD, PD in small infants, complication of PD, supplemental therapy, the future of PD in children, transplantation of patients on PD.

C. Nutritional Management of Pediatric patients on chronic Dialysis:

Introduction, Nutritional Assessment, Anthropometrics parameters, Biochemical assessment, Radiological assessment, Dietary recommendations, Energy requirements, protein require, Lipid require, sodium, Potassium, water, renal osteodystrophy, Vit D therapy, vitamins, Trace elements: Iron, Zinc and Copper, Nutritional considerations for the infant receiving CAPD or CCPD treatment.

D. Psychosocial problems related to dialysis in pediatric patients:

Introduction, Adjustment, compliance, neuropsychological development, rehabilitation.

E. Pediatric CAV Hemofiltration:

Pediatric operational principles of CAVH, characteristics of available hemofilters, practical operational details, clinical experience in the neonate, clinical experience in older children, conclusion.

F. Preemptive Renal Transplantation in infancy: -

Introduction, indications for renal replacement therapy in infancy, renal failure in infancy, growth in infants with renal failure, neurologic development in infants with renal failure, dialysis in infants, PD in infancy, complication of PD in infancy, immunologic effects of PD in the infant hemodialysis in infancy, Renal transplantation in infancy, preparation for infant transplant, post transplantation complication, immunosuppression protocols, identification and treatment of allograft rejection, growth post-transplant in the infant, cost and (RE) hospitalization in the infant with ESRD, Summary.

G. Management of common Electrolyte Disorders in children.

H. Clinical Care Coordinator: The Pediatric Nephrology Technologist of the Future.

5.Recent Advance in Dialysis and Nephrology

a. Artificial Intelligence in Renal Sciences and Dialysis Technology
b. Machine Learning Models
c. Connected neural network
d. Convolutional neural network (CNN)
e. Recurrent neural network (RNN)
f. Generative adversarial network (GAN), and deep reinforcement learning

6.Advanced HD procedures

a. Nocturnal Dialysis
b. Online dialysis
c. Daily dialysis
d. Telemedicine in Dialysis Practice

7.Calculators

a. IV rate and dose
b. Access recirculation
c. URR
d. Concentration
e. GFR Adult/Pedi
f. Kt/V (different formulas)
g. Urea volume distribution
h. Creatinine Clearance
i. BMI
j. Weight and measures (Length, volume, weight, Fahrenheit, Celsius etc)

8.Cyber Nephrology

Renal transplantation and coordination & psychology (60 Hours)

A. Renal transplantation and coordination

1. History of Transplantation
2. Characteristics of the allogenic immune response
3. Tolerance and immunity:
    a. Self – non-self-discrimination
    b. Antigen recognition
    c. Immune tolerance

4.Transplantation antigens:

    a. ABO, Monocyte and Endothelial cells Ag
    b. Major + Minor Histocompatibility Ag

5.Major Histocompatibility Complex

    a. HLA GI and HLA GII
    b. Nature of Allorecognition
    c. Inheritance of HLA.

6.Tissue typing:

    a. HLA typing, short term vs long term/ quality of typing
    b. Matching for split Ags, relative strengths of HLS cocci
    c. Effects of blood transfusion

7.Regulation of the Immune response –

    a. Role of Ag prescribing cell.
    b. T Cell receptor recognition of Ag
    c. CD4/CD8 cells subsets
    d. Accessory molecules
    e. T Cell activation
    f. T Cell energy
    g. Lymphokines and lymphokine receptors
    h. Th1 + Th2 cell subsets

8.Graft rejection

a. Hyperacute / acute/ accelerated/chronic
b. Mechanisms – Ab mediated/T cell mediated/ Delayed Type/ hypersensitivity – mediated NK cell mediated.

9.Mechanisms of Immunosuppression – Corticosteroids/ Azathioprine/FK506/ Rapamycin/Polyclonal immune Globulins/MAb

10. Donor specific immune tolerance/ Tolerance induction by blockade of co stimulation

11. Evaluation of the donor + recipient – special issued + consideration prior kidney Transplantation /Age/Diabetes mellitus/cardiovascular disease/ infections/Malignant neoplasms / metabolic bone disease GI disease/ pulmonary/ urologic evaluation/ systemic disease /psychiatric problems/ vascular disease.

12. Immunological evaluation of the Transplant recipient – typing + Ag matching

13. Screening of Humoral sensitization

14. Cross matching – techniques

15. ABO Blood group matching/ family testing to determine haplotypes/ Cellular assays for HLA testing/ Analysis of survival data.

16. Kidney donation – live donation – non related / related donors, cadaver.

a. Cadaver organ harvesting and preservation
b. Kidney preservation – solutions

17.Transplant surgery + potential complications – Pre-OP care/Surgical technique, post OP management/ potential complications.

18.Immunosuppressive therapy – Induction protocols/ maintenance protocols. AZA/ Steroids/ CSA – Pharmacology – drug interactions

19.Antirejection therapy – Pulse corticosteroids / ALS / OKT3

20.Clinical Approach to Allograft dysfunction – ATN/ CSA/hyperacute rejection

a. Ace or Acute rejection / other causes of DGF/ late chronic dysfunction.

21.Pathological diagnosis of Allograft dysfunction

22.Recurrent Glomerulonephritis

23.Denovo injury

24.Medical complications – Infections disease – Time table – viral infections – CMV/EBV

a. Bacterial Infection– UTI – Opp. Bact. Infections/ Fungal Infection
b. Cardiovascular disease – HT
c. Lipid disorders

d. Liver disease – HBV/HCV

e. Malignancy

f. M,

g. Diabetes Mellitus, hematological problems

25.Pancreatic – Kidney transplantation

26.Pediatric transplantation – problems.

B. Psychology

1. Psychiatric Assessment
    a. Introduction to clinical assessment
    b. The Psychiatric interview
    c. The mental status Exam
    d. Physical Exam and laboratory evaluation
    e. Intelligence testing and neuropsychological assessment
    f. Personality assessment
2. Psychiatric Aspects of Renal Care
    a. Introduction
    b. Psychiatric problems: uncooperativeness-cause of uncooperativeness
    c. Depression and Suicide, anxiety, rehabilitation problems, sexual problems
    d. Psychiatric fitness for Transplantation
    e. Treatment of psychiatric problems- preventive therapy-group therapies-environmental Manipulations-psychotherapy-pharmacotherapy-behavioral sexual techniques
3. Rehabilitation and Psychosocial Issues
    a. Endurance Exercise training in hemodialysis patients
    b. Psychosocial rehabilitation of Adult dialysis patients
    c. Ethical Dilemmas in Dialysis: To initiate or withdraw Therapy
4. Mental Disorders
    a. Diagnosis and classification in psychiatry
    b. Psychiatry history and examination
    c. Organic (Including Symptomatic) mental disorders
    d. Psychoactive substance uses disorders
    e. Schizophrenia
    f. Mood Disorders, Anxiety and personality
    g. Other psychotic Disorders
    h. Neurotic, stress-related and somatoform disorders
    i. Disorders of Adult personality and behavior
    j. Sexual disorders
    k. Sleep disorders

l. Behavioral syndromes associated with psychological disturbances and physiological factors

m. Mental retardation

n. Child psychiatry

o. Psychopharmacology

p. Biological methods of treatment

q. Psychoanalysis

r. Psycho treatment

s. Emergency psychiatry

t. Legal and ethical issues in psychiatry

u. Common psychiatry

QUALITY ASSURANCE AND MANAGEMENT OF DIALYSIS UNIT (60 Hours)

Introduction

1. The Basics of Quality Assurance
    a. DEFINITION
    b. QUALITY ASSURANCE METHODS
    c. The JCAHO 10-Step Process for Monitoring and Evaluation
    d. Comprehensive Quality Assurance Program
    e. Quality Control
    f. Continuous Improvement Concepts
2. Water Treatment
    a. RISKS AND HAZARDS
    b. QUALITY ASSURANCE FOR WATER TREATMENT SYSTEMS
    c. Policies and Procedures.
    d. Staff Training and Education
    e. Monitoring and Evaluation.
    f. Daily Monitoring
    g. Monthly Monitoring
    h. Patient Monitoring
    i. Home Dialysis Monitoring

j. Other Monitoring

k. Prevention

3. Dialysis Delivery System
    a. Technical description of device
    b. Risks and hazard
    c. Quality assurance for dialysis delivery systems
    d. policies and procedures
    e. staff training and education
    f. monitoring and evaluation
        1. Daily monitoring‟s
        2. Monthly monitoring‟s
        3. Patient monitoring
        4. Home dialysis monitoring
        5. Other monitoring
        6. Prevention
4. Dialysate and Dialysate Concentrate
5. Hemodialyzers
6. Ancillary Devices and Equipment
7. Anticoagulation
8. Vascular Access Devices
9. Hemodialyzer Reuse
10. Infection Control
11. Handling of Toxic Chemicals
12. Basics of Management of health care organization include administrative aspects of dialysis unit
13. Patient education & health patient with RRT
    - Patient Education
    - Communication
    - Patient communication problems
    - Explanation of examinations
    - Interacting with terminally ill patient

- Informed Consent
- Patient counseling on
    - Living with CKD
    - Life with HD and CAPD
    - Living with a transplanted kidney.
- Evolution of Quality management
- Quality assurance methods
- Patient satisfaction
- Standard operating procedure
- Quality certification
- Accreditation to national agencies

Advanced Renal Replacement Therapy Practical's (30 Hours)

Hemodialysis

1. Alternative Chronic Hemodialysis Prescriptions
    - Daily Dialysis and Outcome
    - Nocturnal hemodialysis
    - Short daily hemodialysis
    - Outcome studies
2. Alternative approaches to quantification of Dialysis
    - Urea reduction Ration and solute removal index
    - Standard measure of the Dose of Dialysis: Kt/V
3. Management of the maintenance‟s hemodialysis patient
    - Uremic syndrome
    - Anemia
    - Cardiovascular disease
    - Vascular calcification
    - Calciphylaxis
    - Nutrition
    - Infection and immunity
4. Plasmapheresis
5. MARS
6. The future of Renal Replacement Therapy
7. Peritoneal Dialysis

a. Peritoneal dialysis in special populations

i. Rapid transporters

ii. Elderly patients

iii. Patients with diabetes mellitus

iv. Black and Hispanic patients‟

v. Pediatric patients

vi. Patients infected with the human immunodeficiency Virus.

vii. Pregnant patients

b.Peritoneal dialysis in a Nonchronic setting.

i. Acute renal failure

ii. Nonnumeric uses of peritoneal dialysis

c.Management of Complications of peritoneal dialysis

i. Cardiovascular disease

ii. Comparison of hemodialysis and peritoneal dialysis

iii. Indications for switching from peritoneal dialysis to hemodialysis.

iv. Noninfectious complications

v. Infections complications

d.Future trends in peritoneal dialysis

i. Mini PET

8. Pediatric Dialysis

- Pediatric HD
- Pediatric PD
- Nutritional Management of Pediatric patients on chronic Dialysis:
- Pediatric CAV Hemofiltration:
- Management of common Electrolyte Disorders in children.
- Clinical Care Coordinator: The Pediatric Nephrology Technologist of the Future.

9. Recent Advance in Dialysis and Nephrology

- Artificial Intelligence in Renal Sciences and Dialysis Technology
- Machine Learning Models
- Connected neural network

- Convolutional neural network (CNN)
- Recurrent neural network (RNN)
- Generative adversarial network (GAN), and deep reinforcement learning

10. Advanced HD procedures
    - Nocturnal Dialysis
    - Online dialysis
    - Daily dialysis
    - Telemedicine in Dialysis Practice
11. Calculators
    - IV rate and dose
    - Access recirculation
    - URR
    - Concentration
    - GFR Adult/Pedi
    - Kt/V (different formulas)
    - Urea volume distribution
    - Creatinine Clearance
    - BMI
    - Weight and measures (Length, volume, weight, Fahrenheit, Celsius etc)
12. Cyber Nephrology

Renal transplantation and coordination & psychology Practical's (30 Hours)

1. Renal transplantation and coordination

2. Tissue typing:

3. Screening of Humoral sensitization

4. Cross matching

5. ABO Blood group matching/ family testing to determine haplotypes/ Cellular assays for HLA testing/ Analysis of survival data

6. Kidney donation – live donation – non related / related donors, cadaver

7. Clinical Approach to Allograft dysfunction

8. Psychiatric Assessment

9. Rehabilitation and Psychosocial Issues

10. Grief Counselling

Quality assurance and management of Dialysis unit Practical's (30 Hours)

1. QUALITY ASSURANCE METHODS

2. JCAHO 10-Step Process

3. Quality Control

4. QUALITY ASSURANCE FOR WATER TREATMENT SYSTEMS

5. Quality certification

6. Accreditation to national agencies

Residency part – IV (180 Hours)

In the residency the professional is expected to work and contribute in the Nephrology OPD, Nephrology Ward, ICUs & Dialysis Therapy unit.

Project / Thesis/ Dissertation (180 Hours)

Each candidate will have to carry out of a dissertation on the related subject. The dissertation will be guided by one or two members of the faculty of the department. The dissertation will be evaluated by the External/Internal Examiners. The final dissertation duly approved by the External/Internal examiners will be submitted to the Dean‟s office with the result.

Competency

Professional conduct: The Masters of Dialysis Therapy is specifically aimed at those pursuing a professional career in Dialysis Therapy. It is designed to provide specialized training both in basic scientific principles of modern Dialysis and in the application of these principles to the understanding of a wide variety of renal-uro disorders. It is designed as a higher degree program suitable for graduates having some experience in Dialysis Therapy. It is designed to provide training for a future career in research in basic and clinical renal science. The program aims to enhance the scientific skills of clinicians and to provide non-clinical graduates with an insight into clinical problems that will allow them to work alongside clinicians in clinical research projects.

PERFORMANCE CRITERIA	**INDICATOR**			**Measurement Criteria**
	Knowledge	***Skill***	***Behaviour***	
1. Ability to understand the theory related to Dialysis therapy.	Basic knowledge of all the subjects including human Anatomy, Physiology, Dialysis Therapies, Recent Advances, Transplantation, Pharmacology, Pathology, Microbiology, Biochemistry, Ethics & Pedagogy, Applied DT etc.	Should be able to present seminars and demonstrate facilitation of adult teaching, & tutorials related to the syllabus. Should be able to educate UG-level students.	Should be able to behave well with the teachers and students. 2. Should be soft-spoken. 3.Demonstrate good communication & comprehensive proficiency.	Class tests (debates, essay questions, MCQs, etc.) 2.Summative assessment 3.Formative assessment 4. Pedagogy 5.Seminar presentation 6.Case presentation 7.Self-assessment & Peer-assessment 8.Viva voce 9.Patient

				simulation 10.Student projects 11.Problem & Cases Based Learning
2.Ability to understand the roles and responsibilities of a Dialysis therapist	1. In-depth Knowledge of the principles of dialysis. 2. In-depth knowledge of all the modalities as well as procedures of dialysis therapy treatment. 3.In-depth knowledge of all the complications, their preventive measures, management as well & treatment options. 4.Adequate knowledge of managerial responsibilitie s & roles of a clinical supervisor. 5.Adequate knowledge of logistics & inventory	1. Demonstrate an ideal supervisory/ managerial ability to guide & lead the dialysis team. 2.Demonstrate inventory & logistical management. 3.Ability to decide on appropriate dialysis prescription. 4.Perform all the processes of initiating & terminating dialysis therapy. 5.Identify the complications during dialysis and provide instant management as well as bedside teaching of the same to the UG students, interns & other trainees.	1.Have a good working support with the dialysis team & the students. 2.Greet & respect patient & their attendants with care & appropriate manners. 3.Provide counselling to the patient as appropriate .	1.Class tests (debates, essay questions, MCQs, etc.) 2.Summative assessment 3.Formative assessment 4.OSCE, OSPE 5.Clinical posting & logbooks 6.Seminar presentation 7.Case presentation 8.Self-assessment & Peer-assessment 9.Problem-based learning exercises 10.Viva voce 11.Patient simulation 12.Student projects

	management of dialysis therapy.			13.Problem & Cases Based Learning
3. Ability to consider a patient‟s health as their utmost priority.	1. Understand & explain the physiological condition of a dialysis patient. 2. Understand & explain the practice of medical sociology & psychology in dialysis therapy.	1. Identify the patient‟s needs & requirements based on their clinical presentation & lab workups. 2.Explain Renal Replacement Therapy options & demonstrate patient education to help them make the optimum choice of therapy. 3.Plan the prescription of dialysis therapy. Identify & plan a dynamic patient-tailored renal diet consultation & plan renal diet/nutrition.	1. Greet & respect patient & their attendants with care & appropriate manners. 2.Understand & empathize with a patient‟s psychologi cal & social issues while constructiv ely communic ating with them. 3.Ensure equal care & treatment are given to all patients. 4.Respect patients regarding their care plan decisions.	1.Class tests (debates, essay questions, MCQs, etc.) 2.Summative assessment 3.Formative assessment 4.OSCE, OSPE 5.Clinical posting & logbooks 6.Seminar presentation 7.Case presentation 8.Self-assessment & Peer-assessment 9.Problem-based learning exercises 10.Viva voce 11.Patient simulation 12.Student projects 13.Problem & Cases Based Learning

4. Ability to perform different modalities of Dialysis.	1. Should have knowledge of all the extracorporeal therapies including- Haemodialysis, Peritoneal dialysis, Plasmapheresis, CRRT modalities, Hemoperfusion, SPAD, MARS etc.	1.Perform all the aspects of therapeutic & clinical practice pertaining to haemodialysis and peritoneal dialysis. 2. Demonstrate efficient patient as well as their attendant/fa mily education & counselling pertaining to HD & PD procedures, lifestyle changes, diet restrictions, vascular access care, etc. 3.Able to perform and demonstrate the priming of plasmapheresis to all the UG students. 4.Perform hemoperfusion and management of the complications during the treatment. 5.Can present or demonstrate all the therapies to the UG students and another trainee.	1.Demonstrate adaptability to different situations & environme nts at the workplace. 2. Demonstra te good working conduct under stressful situations.	1. Class tests (debates, essay questions, MCQs, etc.) 2.Summative assessment 3.Formative assessment 4.OSCE, OSPE 5.Clinical posting & logbooks 6.Seminar presentation 7.Case presentation 8.Self-assessment & Peer-assessment 9.Problem-based learning exercises 10.Viva voce 11.Patient simulation 12.Student projects 13.Problem & Cases Based Learning

5. Ability to independently deliver the best quality dialysis therapies with excellent clinical competency.	1. Comprehend & explain the principles of renal replacement therapies & each modality. 2. Comprehend & explain the principles & modalities of extracorporea l blood therapies. 3. Comprehend & explain the concept of renal transplantatio n & coordination (both pre & post transplantatio n). 4. Comprehend & explain the complications of dialysis therapies their management & treatment. 5. Understand & describe the concept of biomedical instrumentati ons/electroni cs used in dialysis therapies. 6. Understand & describe the basic nursing care in dialysis therapies to provide total patient care. 7. Understand & describe the concept of	1. Independently demonstrate & perform haemodialysis, peritoneal dialysis, and extracorporeal blood therapies. 2. Perform dialysis therapies & extracorporeal blood therapies in critically ill patients in different types of ICUs-Surgical, Medical, Isolation, Cardio-thoracic, Paediatric, Neuro, Coronary Care ICU, etc. 3. Perform dialysis therapy & extracorporeal blood therapies for infectious patients-Covid-19, H1N1, Viro-positive, MRSAs, TB, etc. 4. Perform dialysis therapies & extracorporeal blood therapies in special cases-Paediatric, geriatric, autistic, pregnant women,	1. Demonstrate good interpersonal relationships with the dialysis team. 2. Demonstrate critical thinking & conflict management. 3. Demonstrate leadership & navigation qualities at the workplace.	1.Class tests (debates, essay questions, MCQs, etc.) 2.Summative assessment 3.Formative assessment 4.OSCE, OSPE 5.Clinical posting & logbooks 6.Seminar presentation 7.Case presentation 8.Self-assessment & Peer-assessment 9.Problem-based learning exercises 10.Viva voce 11.Patient simulation 12.Student projects 13.Problem & Cases Based Learning

	research methodology. Emphasize & understand the recent advances in dialysis therapies & the utilization of AI in dialysis therapies.	differently abled patients, etc. 5. Independently demonstrate & perform AVF/AVG cannulation & Central Venous catheter handling. 6. Operate routine maintenance, identification of a malfunction, troubleshootin g, and minor repair for all equipment used in dialysis units such as hemodialysis machines, water treatment plants, dialyzer reprocessing machines, etc. 7. Identify & handle complications of dialysis therapies & extracorporeal therapies of each modality. 8. Demonstrate basic renal nursing care in dialysis patients-safe injection		

		practices, blood & blood products transfusion, IV cannula insertion, IV antibiotic administration s, emergency medication administration during intradialytic complications, anticoagulant iv injections, EPO & iron injections, the concept of the crash cart, etc. 9. Demonstrate BLS & ACLS 10. Development & planning of a dialysis unit for a standalone organization or a healthcare set-up. 11. Participate & assist in research activities		
6. Ability to follow quality assurance guidelines and healthcare policies.	Be able to comprehend and articulate hospital infection control policies, quality assurance and patient safety procedures, and fundamental management principles in the	1. Develop efficient SOPs for a dialysis therapy unit. 2. Understand & demonstrate the emergency codes protocols-Code Red, Code Blue, Code Yellow,	1. Proactive attitude and awareness of potential health and safety hazards in the immediate area. 2. Pleasant support for the patients,	Class tests (debates, essay questions, MCQs, etc.) 2.Summative assessment 3.Formative assessment

	healthcare and dialysis industries.	Code Pink, Code Violet, etc. 3. Demonstrate hospital infection control practices.	facilitating access to the dialysis therapists and fostering patient satisfaction .	4.OSCE, OSPE 5.Clinical posting & logbooks 6.Seminar presentation 7.Case presentation 8.Self-assessment & Peer-assessment 9.Problem-based learning exercises 10.Viva voce 11.Patient simulation 12.Student projects 13.Problem & Cases Based Learning
7. Ability to administer medications under the supervision of a nephrologist.	1. Proper knowledge of the drugs used in the dialysis unit. 2.Knowledge of the administration of different drugs through different routes.	1. Should follow International Patient Safety Goals (IPSG 1to 6) 2. Identify patients correctly. 3. Improve effective communication. 4. Improve the safety of high alert medications.	1. Demonstrate honesty & integrity while patient medication handling. 2. Demonstra te good communic ation with the patient party & the nephrologi sts/physici ans. and other stakeholder s 3. Counsel the patient party effectively	1. Class tests (debates, essay questions, MCQs, etc.) 2. Summative assessment 3.Formative assessment 4.OSCE, OSPE 5.Clinical posting & logbooks 6.Seminar presentation 7.Case

		5. Ensure safe surgery. 6. Reduce the risk of health care associated infections. 7. Reduce the risk of patient harm resulting from falls. 8. To provide medication through IV, IM, SC, and oral routes. 9. Ability to perform drug calculations for all administration routes, give legal pharmaceutic als in accordance with hospital policy, and, where necessary, take appropriate corrective action for complications while being supervised. 10.Demonstrate excellent documentatio n ability.	regarding the medication s & probable side effects	presentation 8.Self-assessment & Peer-assessment 9.Problem-based learning exercises 10.Viva voce 11.Patient simulation 12.Student projects 13.Problem & Cases Based Learning
8. Ability to supervise and educate UG/PG students	1. Comprehend & explain the dialysis therapies & all the modalities.	1. To function as a dialysis therapy faculty. 2. To function as a faculty in	1. Polite and presentable 2. Show good communication &	1. Class tests (debates, essay questions, MCQs, etc.)

	2. Comprehend & explain the basic knowledge of the prescription, complications, and treatment. 3. Demonstrate & teach clinical competency & patient care.	facilitating & training students, trainees, and interns with dialysis therapy modalities. 3. Demonstrate & teach students using simulations/dummies for dialysis procedures and vascular access handling. 4. Efficiently perform university/college activities as per the requirements & protocols.	comprehensive proficiency. 3. Maintains good interpersonal relations with the clinical team. 4. Observe professional conduct & integrity as dialysis therapy faculty.	2.Summative assessment 3.Formative assessment 4. Pedagogy 5.Seminar presentation 6.Case presentation 7.Self-assessment & Peer-assessment 8.Patient simulation 9.Student projects 10.Problem & Cases Based Learning

9. Ability to actively exhibit a good approach in the Research field.	1. Good knowledge of the recent trends and studies related to kidney disease patients and dialysis. 2. Good knowledge of the research methodology & statistical theory.	1.Should be able to observe all the protocols & policies of the Institutional Research Committee and Ethical Committee. 2. Should be able to propose & conduct any type of research projects, case reports, article reviews, etc. related to one‟s field/subject. 3. Should be able to engage, guide/co-guide & facilitate all the staff and students to perform their research project. 4. Should be proficiently aware of global studies & latest trends in research. 5. Should be able to facilitate & assist the nephrologists, other healthcare professionals, or other research enthusiasts as & when approached for	1. Demonstrate ethical conduct & confidentiality while handling patient data & information. 2. Respect the study participants ‟ decisions & their involvement. 3. Acknowledge & credit all the members involved in the study. 4. Ability to be a team player. 5. Demonstrate critical thinking & thorough attention to detail. 6. Organizational and communications skills. 7. Motivated & enthusiastic about their work & it‟s completion on their own.	1. Student projects 2.Summative assessment 3.Formative assessment 4.Case presentation 5. Problem & Cases Based Learning

		their studies as deemed. 6. Demonstrate effectively explaining the involvement of the study participants in detail & acquiring informed consent forms from them. 7. Demonstrate professional presentation of their study with good comprehensive proficiency.		